[redacted]

A 1)

a) $\dot{Q}_{aus} = ?$

stationärer Fließprozess im Kessel

$$0 = \dot{m}_{ein}(h_{ein} - h_a) + \dot{Q}_R - \cancel{\dot{W}_t}^{\,0} - \dot{Q}_{aus}$$

$$\dot{Q}_{aus} = \dot{m}_{ein}(h_e - h_a) + \dot{Q}_R = \underline{\underline{62{,}182\ kW}}$$

Tab A-2

$h_e = h_f(70°C) = 292{,}98\ kJ/kg$

$h_a = h_f(100°C) = 419{,}04\ kJ/kg$

b) $$\bar{T} = \frac{\int_e^a T\,ds}{s_a - s_e} = \frac{h_a - h_e}{s_a - s_e} = \frac{126{,}06}{0{,}352} = \underline{\underline{358{,}125\ K}}$$

~~$s_a = s_f(70$~~ Tab A-2

$s_e = s_f(70°C) = 0{,}5545\ kJ/kgK$

$s_a = s_f(100°C) = 1{,}3069\ kJ/kgK$

c)

$$0 = \dot{m}(s_2 - s_1) + \frac{\dot{Q}}{T_G} + S_{erz}$$

$$S_{erz} = \dot{m}(s_1 - s_2) - \frac{\dot{Q}}{T_G} = 0{,}1056 - \frac{Q_{aus}}{T_G} = 0{,}1056 + \frac{61{,}182}{275}$$

$$= 0{,}3169 \text{ kJ/kgK}$$

$$= 316{,}39 \text{ J/kgK}$$

auf zweitem Blatt

d) Halboffenes System

$$\Delta E = m_2 u_2 - m_1 u_1 = \Delta m (h)_{ein} + Q - \cancel{W}^{\,0}$$

Tab A.2

$$(m_1 + \Delta m) u_2 - m_1 u_1 = \Delta m\, h_{ein} + Q$$

$$m_1 u_2 - m_1 u_1 - Q = \Delta m (h_{ein} - u_2)$$

$$\Delta m = \frac{m_1 u_2 - m_1 u_1 - Q}{h_{ein} - u_2} = 572{,}4 \text{ kg}$$

$h_{ein} = h_f(20°C) = 83{,}96$ kJ/kg

$u_1 = u(100°C, x = 0{,}005)$

$= u_f(100°C) + x(u_g - u_f) = 420{,}38$ kJ/kg

$u_2 = u_f(70°) = 292{,}95$ kJ/kg

$Q = 35 \cdot 10^3$ kJ

[redacted]

A1)

e) $\Delta m_{12} = 3600\,kg$

Halbofflas $s_{1,sh}$

$$\Delta S_{12} = m_2 s_2 - m_1 s_1 = (m_1 + \Delta m_{12}) s_2 - m_1 s_1 = \underline{\underline{1237,84\,kJ}}$$

Tab A-2 bei 100°C

$$s_1 = s(100°C, x = 0{,}005) = s_f + x(s_g - s_f) = 7{,}33714\ kJ/kg\,K$$

$$s_2 = s_f(70°) = 0{,}9545$$

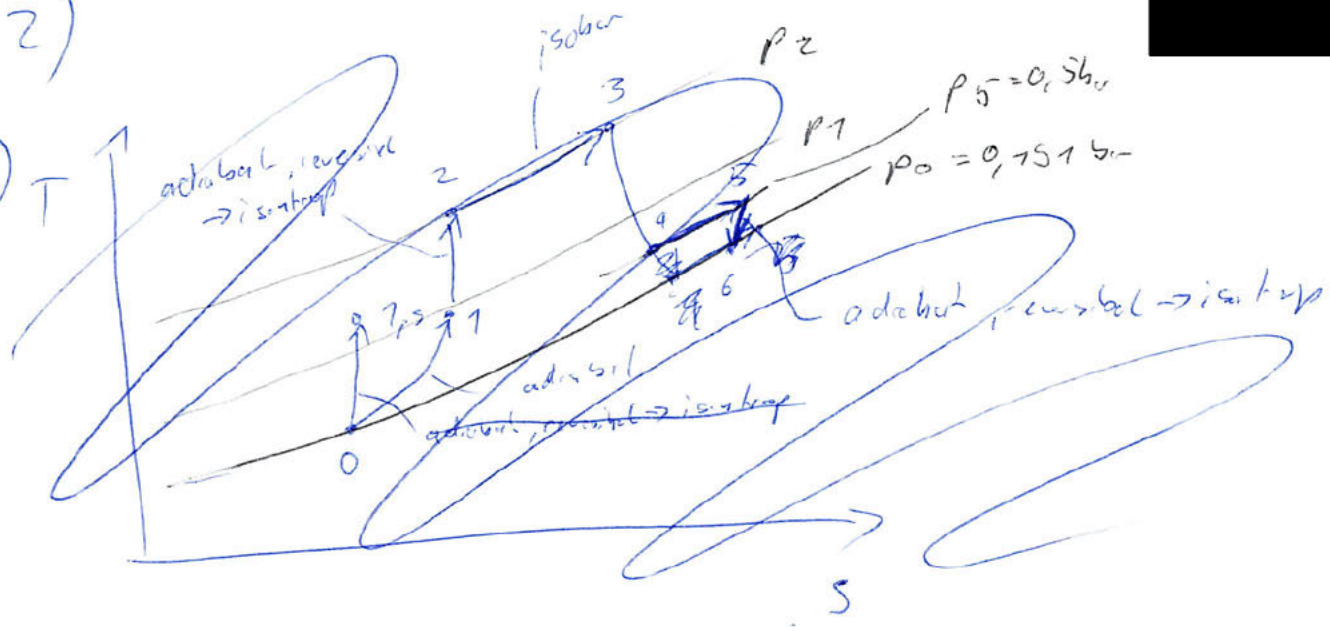
c) $$\bar{T} = \frac{\int_e^a T\,ds}{s_a - s_e} = \frac{\int_e^a T\,ds}{\int_{T_e}^{T_a} \frac{c}{T}\,dT} = \frac{\int_e^a T\,ds}{c \ln\left(\frac{T_a}{T_e}\right)}$$

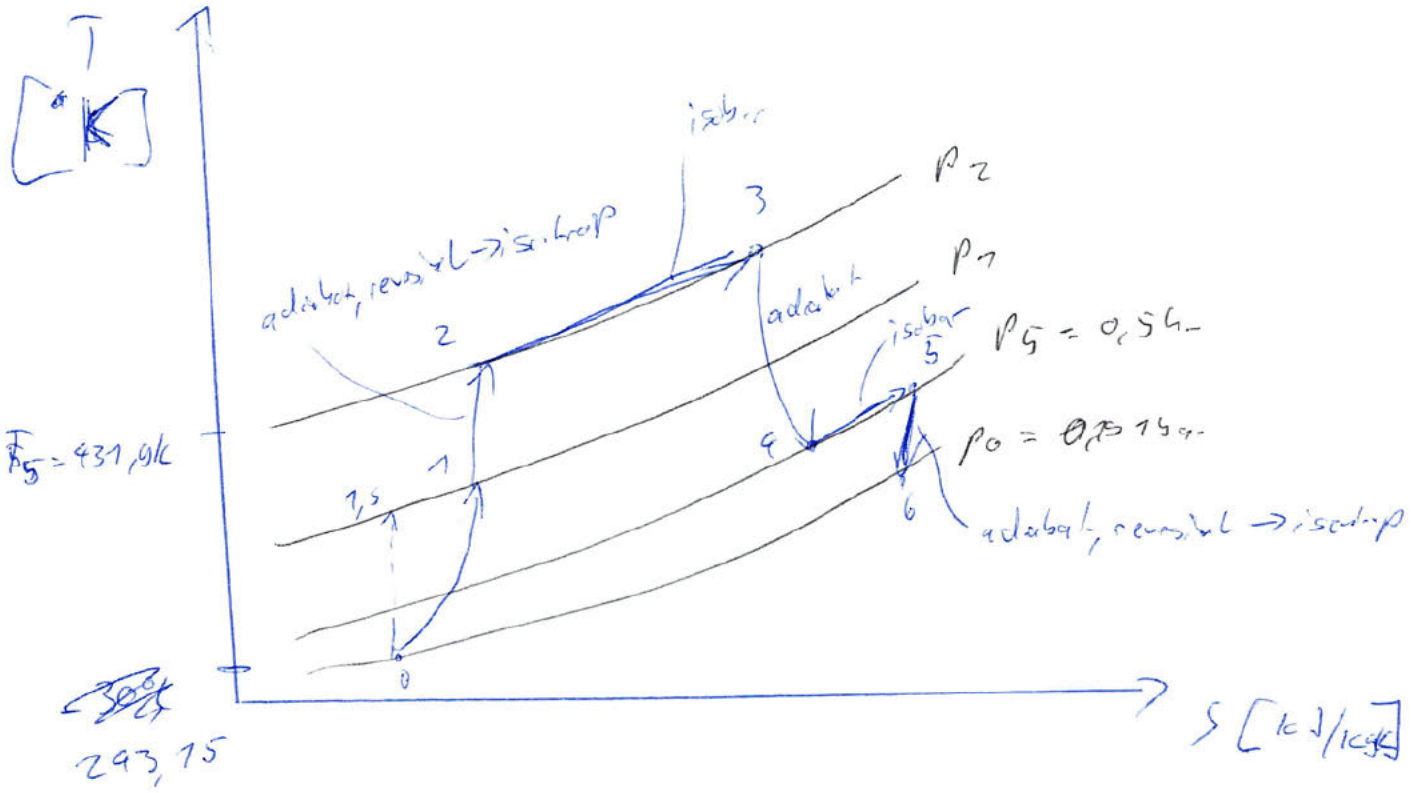
$$= \frac{Q}{c \ln\left(\frac{T_a}{T_e}\right)} = \frac{c(T_a - T_e)}{c \ln\left(\frac{T_a}{T_e}\right)} = \frac{T_a - T_e}{\ln\left(\frac{T_a}{T_e}\right)} =$$

[redacted]

A2)

a)





b) $w_6 = ?$ $T_6 = ?$

$w_5 = 220\,m/s$, $T_5 = 431{,}5\,K$, $p_5 = 0{,}5\,bar$

$p_6 = p_0 = 0{,}1917\,bar$

isentrop $s_5 = s_6$

stationärer Fließprozess

$$0 = \dot{m}\left(h_5 - h_6 + \frac{w_5^2 - w_6^2}{2}\right) + \cancel{\dot{Q}} - \cancel{\dot{W}_t}$$

adiabat

reversibel

$$h_5 - h_6 = c_p(T_5 - T_6) = +146{,}2286\,kJ/kg$$

$$c_p = \kappa \cdot c_v = n \cdot c_v = 1{,}4084\,kJ/kgK$$

$$\Rightarrow R = c_p - c_v = 0{,}4024\,kJ/kgK$$

$$h_5 - h_6 + \frac{w_5^2 - w_6^2}{2} = 0$$

$$2(h_5 - h_6) + w_5^2 = w_6^2$$

$$w_6 = \sqrt{\ldots}$$

$$\cancel{= 219{,}334\,m/s}$$

$$= 1729{,}98\,m/s$$

isentrop $s_5 = s_6$

$$0 = c_p \ln\left(\frac{T_6}{T_5}\right) - R \cdot \ln\left(\frac{p_6}{p_5}\right)$$

$$R \cdot \ln\left(\frac{p_6}{p_5}\right) = c_p \ln\left(\frac{T_6}{T_5}\right) \quad | e^{\wedge}$$

$$e^{R \cdot \ln\left(\frac{p_6}{p_5}\right)} = e^{c_p \ln\left(\frac{T_6}{T_5}\right)}$$

$$\left(\frac{p_6}{p_5}\right)^R = \left(\frac{T_6}{T_5}\right)^{c_p} \rightarrow T_6 = T_5 \left(\frac{p_6}{p_5}\right)^{\frac{R}{c_p}} = 328{,}075\,K$$

[redacted]

c)

$$\Delta e_{x,str} = e_{x,str\,6} - e_{x\,str\,0} = h_6 - h_0 - T_0(s_6 - s_0) + \frac{w_6^2}{2} - \frac{w_0^2}{2}$$

$$h_6 - h_0 = c_p (T_6 - T_0) = 119{,}608\ kJ/kg$$

$$s_6 - s_0 = c_p \ln\left(\frac{T_6}{T_0}\right) - R \ln\left(\frac{p_6}{p_0}\right) = 0{,}421905\ kJ/kg$$

$$\frac{w_0^2}{2} = 20\ kJ/kg$$

$$\frac{w_6^2}{2} = 1487{,}8\ kJ/kg$$

$$\Delta e_{x,str} = 1495{,}21\ kW/kg$$

d)

$$\dot{E}_{x\,vol} = T_0 \cdot \dot{S}_{irr}$$

stationäre Fließprozesse

$$0 = \dot{m}(s_0 - s_6) + \frac{\dot{Q}}{T_G} + \dot{S}_{irr}$$

$$s_{irr} = \overbrace{s_6 - s_0}^{c} - \frac{q}{T_G} = 0{,}421905\ kJ/kg - \frac{1195}{1285}\ kJ/kg$$

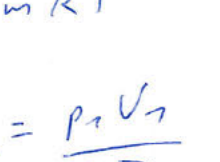
$$= 0{,}505\ kJ/kg$$

$$e_{x,vol} = T_0 \cdot s_{irr} = 243{,}15 \cdot 0{,}505 = 122{,}832\ kW/kg$$

[redacted]

A 3)

a) $R = \frac{\overline{R}}{M} = \frac{8314}{50} = 166{,}28\ J/kg$



$p_{EW} = p_0 + F \cdot A = 1\,bar + m \cdot g \cdot \pi \left(\frac{D}{2}\right)^2 = 1{,}247\ bar$

$p_{g,1} = p_G = p_{EW} = \underline{\underline{1{,}247\ bar}}$

perfektes Gas:

$V_1 = 3{,}14 \cdot 10^{-3}\ m^3$

$T_1 = 773{,}15\ K$

$pV = mRT$

$m = \frac{p_1 V_1}{R T_1} = \underline{\underline{0{,}0312\ kg}}$

b) Da sich an den ungestörten Bedingungen nichts ändert, drücken immer noch 32kg + 1bar atmosphärischen Druck, dadurch ist $p_{g,2} = p_{g,1} = \underline{\underline{1{,}247\ bar}}$.

$\Delta E = 0 = m_{EW}(u_{2g} - u_{1g}) + m_g(u_2 - u_1)$

$= m_{EW}(u_{2g} - u_{1g}) + m_g(c_v(T_2 - T_1))$

c)

$m_1 = m_2$

$$\Delta E = m_2 u_2 - m_1 u_1 = Q - \cancel{W} \to 0$$

$$\cancel{Q = m_{EW}(u_2 - u_1)}$$

$$\cancel{u_1 = u(1{,}247\,bar,\ 0°C) = u(T = 0°C) =}$$ inkompressibel

$$\cancel{u_2 = u(1{,}247\,bar,\ 0{,}003°C) = u(T = 0{,}003°C)}$$

$$\Rightarrow Q = m_g(u_2 - u_1) = m_g c_w (T_2 - T_1) = \underline{\underline{9{,}873\,kJ}}$$

d)

$x_1 = 0{,}6$ $\quad 0°C$

$$u_1 = u(T = 0°C,\ x = 0{,}6) = u_{fest} + x(u_{flüssig} - u_{fest}) = -133{,}4702$$

$\cancel{u_{2G}}$

$$Q = m_{EW}(u_2 - u_1) = \cancel{9{,}873\,kJ}$$

$$u_2 = \frac{Q}{m_{EW}} + u_1 = 98{,}73\,kJ/kg + (-133{,}4702)\,kJ/kg = -34{,}6802\,kJ/kg$$

$T_2 = 0{,}003\,°C$

$$x = \frac{u_2 - u_{fest}}{u_{flüssig} - u_{fest}} = \underline{\underline{0{,}896081}}$$

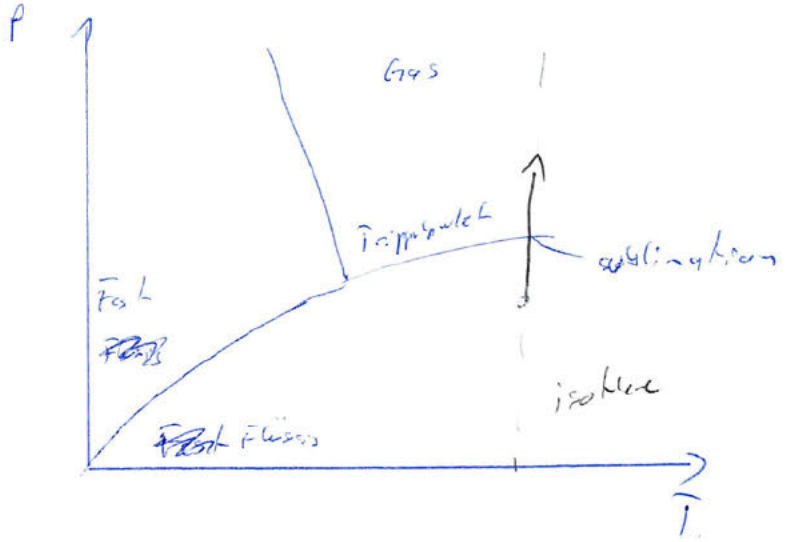
[redacted]

A4)

a) ~~$T_{tripel} = 0°C$~~

$T_{tripel} = 0°C$ $\quad$ $T_{subl}(1\,bar) = -20°C$

somit liegt T_i mit 10K über T_{subl} bei $\underline{T_i = -10°C}$



b) stationärer Fliessprozess bzw verdampfen

0 adiabat

$$0 = \dot{m}(h_2 - h_3) + \cancel{\dot{Q}} - \dot{W}_t$$

$$\dot{W}_k = \dot{m}(h_2 - h_3) \rightarrow \dot{m}_{134} = \frac{\dot{W}_k}{|h_2 - h_3|} = \frac{28 \cdot 10^{-3}}{|237{,}79 - 277{,}31|}$$

$1 \rightarrow 2$ isobare Verdampfung $\qquad = 8{,}39 \cdot 10^{-4}\ kg/s$

$T_v = T_i - 6 = -16°C = T_2$ $\qquad = \underline{0{,}839\ g/s}$

Tab A-11

$h_2 = h(x_2 = 1, T_2 = -16°C) = 237{,}79\ kJ/kg$

adiabat reversibel $3 \to 4 \Rightarrow$ isentrop $s_3 = s_4 = s_g(-16°C) = 0{,}9298$ Tab A-11

$p_3 = 8$ bar

Tab A-12 8 bar

$$h_3 = h(8\,\text{bar}, s = 0{,}9298) = h_{sat} + \frac{0{,}9298 - 0{,}5379}{0{,}9066 - 0{,}5379}(h_{g0} - h_{sat}) = 271{,}314\ \text{kJ/kg}$$

c) $x_4 = 0$ $p_3 = p_4 = 8$ bar

Tab A 11

$s_4 = s_1 = s_f(8\,\text{bar}) = 0{,}3459$

$p_2 = 1{,}5748$ bar ← abgelesen aus Tab A-10 bei $T_2 = -16°C$

$p_1 = p_2 = 1{,}5709$ bar, da isobar

Tab A-11 interpolieren

p	s_f	s_g	h_f	h_g
1,4	0,1035	0,5372	29,77	236,89
1,5709	0,1194	0,529894	29,1945	237,6882
1,6	0,1211	0,5295	29,78	237,97

$$x_1 = \frac{s - s_f}{s_g - s_f} = \underline{\underline{0{,}2795}}$$

[redacted]

A4)

d) $\varepsilon_{IK} = \frac{Q_{ZU}}{Q \cdot W_t} = \frac{\dot{Q}_{IK}}{W_t} = \frac{129,3}{28} = \underline{\underline{4,476}}$

bei Kondensator

$\dot{Q}_{IK}$

$0 = \dot{m}(h_1 - h_2) + \dot{Q} - \cancel{\dot{W}_t}^{\,0}$

$\dot{Q}_{IK} = \dot{m}(h_2 - h_1) = 0,12536\,kJ/s = 125,3\,W$

aufgabe c

$h_1 = h(1,5\,kPa, x = 0,2799) = h_f + x(h_g - h_f) = 87,4684$

$h_2 = 237,29\,kJ/kg$ ← Aufgabe b

e) Die Temperatur im Innern würde isochor weiter fallen.